








**Summer Internship Report**

Thammasat English Program of Engineering: TEPE

Faculty of Engineering, Thammasat University

Academic Year 2021

**Airports of Thailand Public Company Limited**

333 Choet Wutthakat Road, Si Kan, Don Mueang, Bangkok 10210, Thailand

Name of Student: Chanoudom Tann

ID. No. 6610755487

Major: Electrical and Data Engineering Year: 2026

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## Preface

This report is submitted as part of my summer internship program at Thammasat University. It summarizes my practical training experience at **Airports of Thailand Public Company Limited (AOT)** within the **Technology and Communications Strategy Department (**ฝ่ายกลยุทธ์เทคโนโลยีและการสื่อสาร) from 01 June 2026 to 31 July 2026. The goal of this internship was to apply my academic knowledge to real-world operations in aviation technology and communications, gain hands-on professional experience, and learn from industry experts.

I would like to thank **Airports of Thailand PLC (AOT)** for providing me with this valuable opportunity. Special thanks go to my supervisor and mentors in the Technology and Communications Strategy Department, as well as my faculty advisor, for their continuous guidance and support. I am also grateful to my colleagues, family, and friends for their encouragement throughout this journey.

I hope this report serves as a useful reference for future students interested in airport technology and corporate strategy.

## Acknowledgement

This internship report would not have been successfully completed without the invaluable support, guidance, and encouragement of many individuals and organizations.

First and foremost, I would like to express my deepest gratitude to **Airports of Thailand Public Company Limited (AOT)** for providing me with the exceptional opportunity to undertake my summer internship within the **Technology and Communications Strategy Department (**ฝ่ายกลยุทธ์เทคโนโลยีและการสื่อสาร). The experience of working in a leading corporate environment within the aviation sector has been truly transformative for my professional growth.

I am especially indebted to my corporate supervisor and mentors within the Technology and Communications Strategy Department. Their patience, expert knowledge, and willingness to share practical insights greatly enriched my daily learning experience and helped me bridge the gap between academic concepts and real-world execution.

I would also like to express my sincere appreciation to my faculty advisor, Asst.Pro.Dr. Phisan Kaewprapha and my professors at Thammasat University for their continuous academic guidance, encouragement, and support throughout my studies and this internship journey.

Finally, I am grateful to my colleagues at AOT for their welcoming cooperation, as well as to my family and friends for their constant motivation and belief in me.

Thank you all for making this summer internship an unforgettable and rewarding experience.

## Organization Introductions

### I. Background and Business Overview

**Airports of Thailand Public Company Limited (AOT)** is a state-owned enterprise operating under the Ministry of Transport, with the Ministry of Finance holding the majority stake. Listed on the Stock Exchange of Thailand (SET), AOT is entrusted with the vital national responsibility of managing, operating, and developing Thailand's primary commercial international gateways.

AOT currently governs **six major international airports** that serve as critical hubs for global tourism, trade, and logistics:

- **Suvarnabhumi Airport (BKK):** The main international gateway handling long-haul and high-volume passenger traffic.
- **Don Mueang International Airport (DMK):** A historical and major hub primarily focused on regional and low-cost carrier services.
- **Phuket International Airport (HKT):** A crucial regional hub supporting tourism in the southern economic corridor.
- **Chiang Mai International Airport (CNX):** The principal aviation hub for Northern Thailand.
- **Hat Yai International Airport (HDY):** Serving the commercial and transit needs of Southern Thailand.
- **Mae Fah Luang-Chiang Rai International Airport (CEI):** Supporting cultural tourism and regional connectivity in the northernmost province.

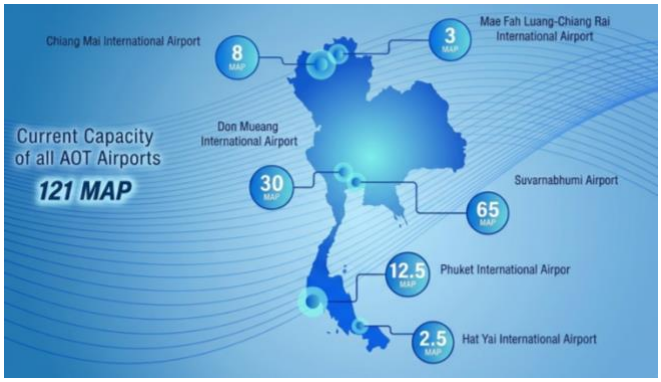



*Figure 1: Locations of AOT's Six International Airports locations of AOT's Six International Airports*

## II. Core Business and Products/Services

AOT's business model extends far beyond basic runway and terminal management. Its revenue streams and service offerings are categorized into two major sectors:

- **Aeronautical Activities:**
    - Landing and parking charges for commercial airlines.
    - Passenger service charges (airport exit fees).
    - Aircraft service equipment and fueling support operations.
- **Non-Aeronautical Activities:**
    - Concession agreements for duty-free shops, retail stores, food and beverage outlets, and commercial advertising within the terminals.
    - Office space rentals, airline lounges, and ground handling services.
    - Property and land development projects surrounding the airport zones to maximize commercial value.

## III. Corporate Vision, Mission, and Core Values

- **Vision:** *"To be a world-class airport operator, focusing on service quality with an emphasis on safety and balanced revenue generation."*
- **Mission:**
    1. To operate and develop airport businesses and related services efficiently and sustainably.
    2. To adhere to international safety and security standards set by organizations like the International Civil Aviation Organization (ICAO).
    3. To support Thailand's economic expansion, national logistics, and tourism industry.
- **Core Values:** Operating with corporate governance, transparency, stakeholder accountability, and sustainable environmental stewardship.

## IV. Organization Policy and Digital Strategy

AOT operates under stringent corporate governance, risk management, and compliance frameworks. To maintain its status as a world-class operator, AOT has heavily prioritized **Digital Transformation and Innovation**.

Key institutional policies focus on:

- **Smart Airport Initiative:** Transitioning traditional terminal processes into automated, frictionless journeys using modern technology.
- **Sustainability and Green Operations:** Implementing eco-friendly infrastructure and moving toward paperless, energy-efficient workflows.
- **Cybersecurity and Data Privacy:** Ensuring robust IT defense mechanisms to protect sensitive operational data, passenger information, and critical national infrastructure systems.

## V. Department Focus: Technology and Communications Strategy Department

During my summer internship, I was assigned to the **Technology and Communications Strategy Department (ฝ่ายกลยุทธ์เทคโนโลยีและการสื่อสาร)**. This division acts as the brain trust behind AOT's technological modernization.

- **Primary Function:** The department is responsible for formulating long-term IT roadmaps, aligning technological capabilities with AOT's corporate master plan, and designing robust communication frameworks across all six airports.
- **Core Processes and Operations:**
    - **Digital Infrastructure Planning:** Developing strategies for enterprise architecture, cloud integration, and enterprise-wide data systems.

- **Innovation Deployment:** Overseeing projects related to passenger processing modernization—such as the Common Use Passenger Processing System (CUPPS), self-service check-in kiosks, and Automated Biometric Identification systems.
- **Internal & External Communications:** Standardizing communication networks, telecommunication channels, and secure data exchange protocols between AOT headquarters, airport branches, and external regulatory partners.
- **Strategic Project Management:** Monitoring technology implementation timelines, conducting feasibility studies for emerging tech, and evaluating vendor solutions to ensure high operational reliability and return on investment.




*Figure 2: Organizational Structure highlighting the Technology and Communications Strategy sector within AOT*

## JOB TRAINING DESCRIPTION AND PROCESS

### I. Overview of Internship Scope and Objectives

During my summer internship at Airports of Thailand PLC (AOT) within the **Technology and Communications Strategy Department**, my core assignment focused on the design, development, integration, and optimization of an internal enterprise application using **Microsoft Power Apps**, **SharePoint Lists**, and **Power Automate**.

The main objectives of this training were:

- To apply theoretical knowledge of electrical and data engineering, software architecture, and system design to real-world corporate workflows.
- To understand the digital transformation strategies of a world-class airport operator.
- To gain hands-on experience in low-code/no-code enterprise application development, database structuring, workflow automation, and dashboard report generation.

### II. Phase-by-Phase Training Process and Chronological Activities

*Phase 1: Onboarding, Orientation, and System Familiarization (Weeks 1–2)*

- **Orientation and Corporate Briefing:** Received a comprehensive overview of AOT's corporate structure, department responsibilities, and safety standards. Configured the workstation and set up collaborative workspace protocols.
- **Project Scoping & Introduction:** Attended briefings regarding internal digital tools and met with department mentors (such as P'Miw) and project stakeholders (including Ajarn Phisan) to discuss project objectives, requirements, and expected deliverables for the enterprise tracking application.

- **Airport Site Visit:** Participated in an operational site visit to Suvarnabhumi Airport to observe real-world airport workflows, user environments, and practical challenges faced by staff, helping align system requirements with actual operational needs.

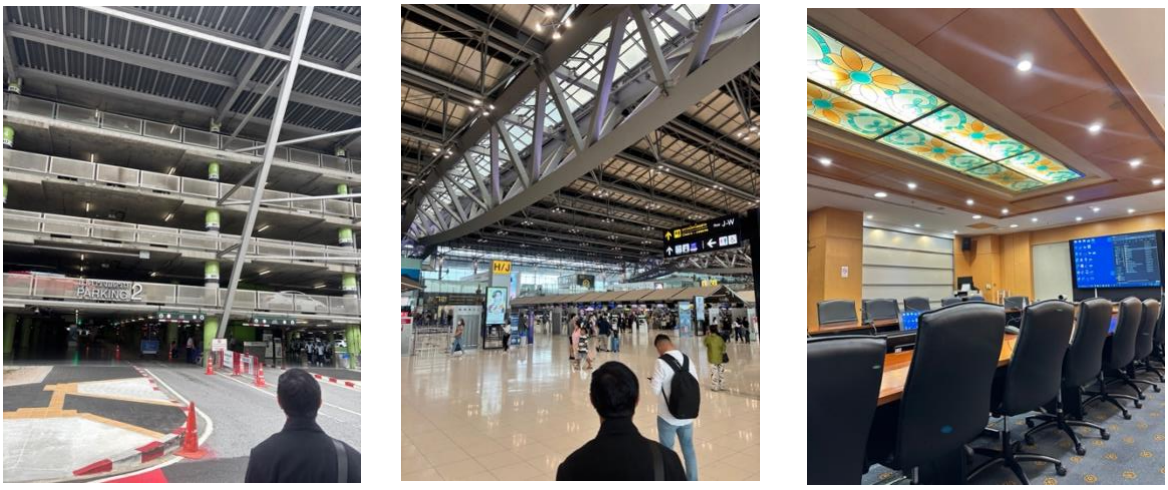


***Figure 3:*** *Operational site visit and corporate activities at Suvarnabhumi Airport, including infrastructure inspection of parking facilities (left), observation of terminal passenger flows (center), and participation in the annual corporate management session (right).*

- **Power Apps Platform Training:** Explored the Power Apps interface, data sources, canvas app architecture, and basic application development concepts.

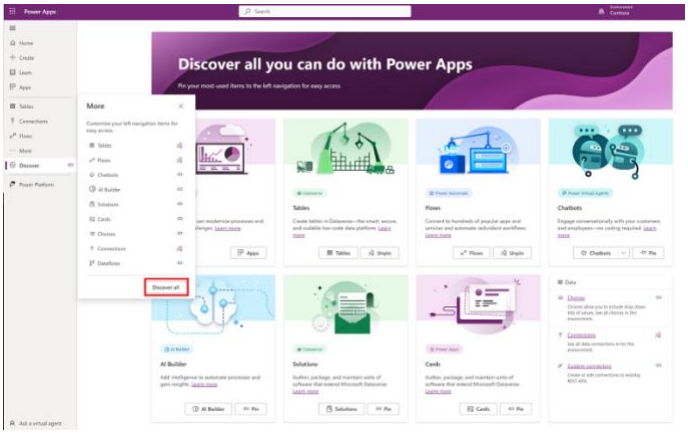



*Figure 4: Overview of the Microsoft Power Apps development environment and available solution types*

*Phase 2: Core Application Development & UI Design (Weeks 3–5)*

- **Login Screen Development:** Designed and configured the initial Login Screen, implementing user authentication navigation, session management, and basic UI controls.
- **Add Application & Form Management:** Developed the data entry forms (Add Application screen), incorporating text inputs, drop-downs, and checkbox functions. Resolved initial data connection bottlenecks and data recording errors.
- **Edit Screen & Record Modification:** Built and refined the Edit Screen to allow users to update, modify, and review existing database entries seamlessly with real-time UI adjustments.
- **Dataset Architecture & SharePoint Integration:** Created, structured, and validated organizational dataset schemas (โครงสร้างฝ่าย). Connected SharePoint Lists as the reliable cloud backend database for application records and verified data mapping accuracy.

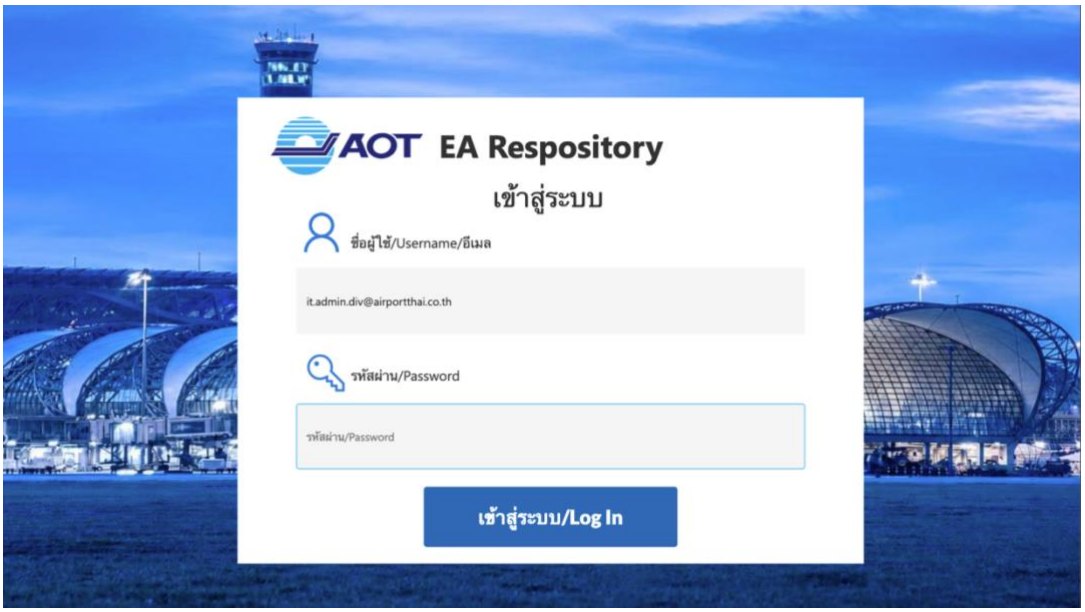



*Figure 5: User authentication and login interface for the EA Respository application*

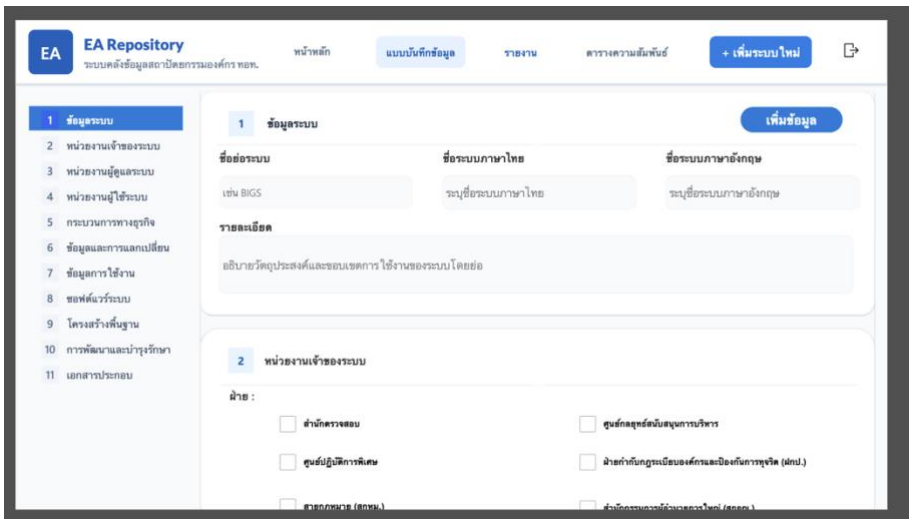



*Figure 6: Data recording and organizational structure input form interface*

## *Phase 3: Workflow Automation & Advanced Features (Weeks 6–7)*

- **Power Automate Integration:** Developed automated workflows using Power Automate to handle system events, such as automated login alerts and scheduled notification triggers.
- **Automated Daily Reporting:** Configured an automated Outlook reporting mechanism to compile and dispatch user activity reports daily at 17:30.
- **Activity Tracking & Error Debugging:** Implemented user tracking features, resolved UI bugs, and optimized form validation based on direct feedback from supervisors (P'Miw).

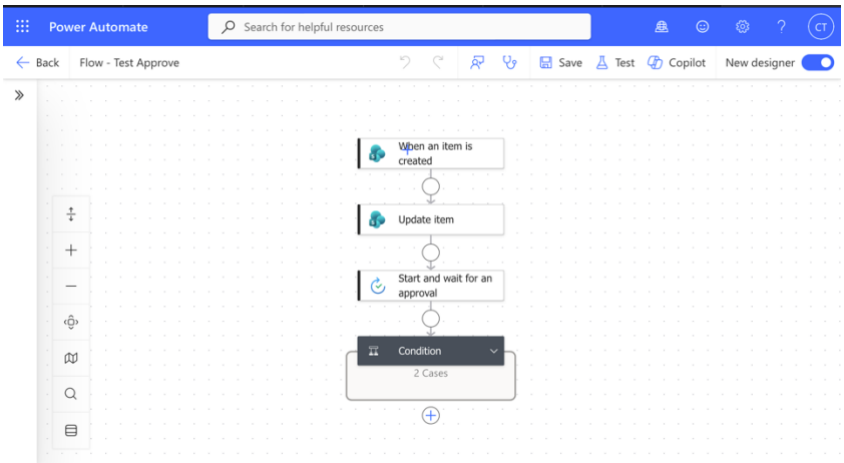



*Figure 7: Automated workflow logic configured in Microsoft Power Automate*

*Phase 4: Dashboard Reports & Enterprise Analytics (Weeks 8–9)*

To provide comprehensive business intelligence and system oversight, multiple analytical report screens and data views were designed and implemented:

- **รายการ (Transaction/Item List) Screen:** Developed data grid views with advanced filtering options for searching and inspecting records.
- **Application Portfolio Report:** Designed reporting views displaying application distribution and filtering metrics.
- **Business Alignment Report:** Connected data sources to visualize how digital projects align with strategic corporate goals.
- **Data & PDPA Report:** Built tracking and compliance visualization modules aligned with data protection standards.
- **Infrastructure & Digital Technology Reports:** Developed technical reporting screens to monitor system health and digital asset inventories.
- **Security / GRC Report:** Configured governance, risk, and compliance metrics tracking.
- **Integration Testing & Refinement:** Conducted cross-module testing to validate data consistency, fix minor UI anomalies, and optimize performance.

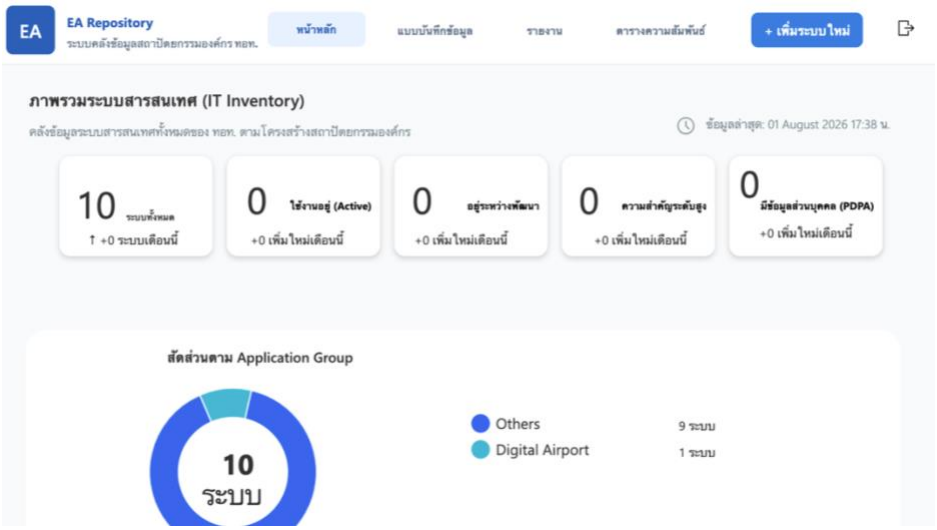



Figure 8: *Main dashboard and IT Inventory overview screen of the EA Repository system*

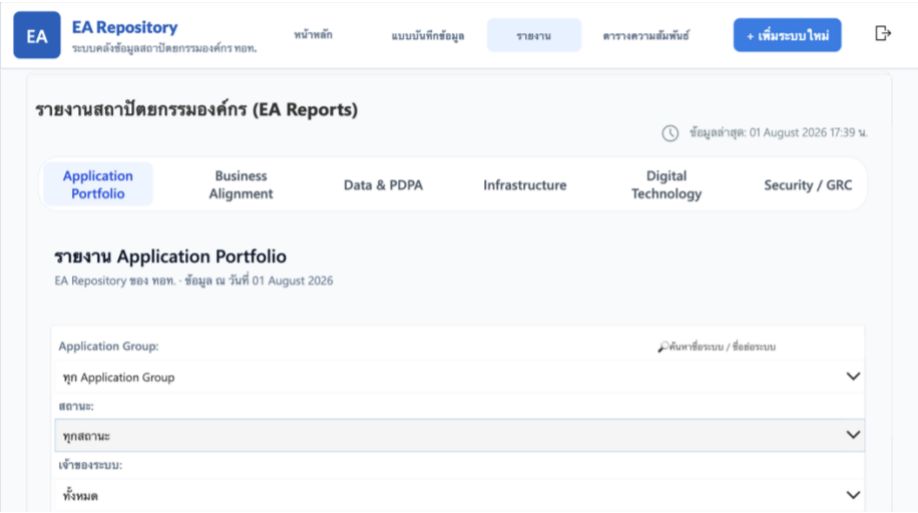



Figure 9: Example of the Application Portfolio reporting dashboard and data filtering module

*Phase 5: Final Polish, Enhancement, and Handover (Week 10)*

- **Home Screen Enhancements:** Added advanced filter tabs, cleaned up layout aesthetics, and refined user experience (UX) elements.
- **Final Project Completion & Documentation:** Completed final system debugging, compiled technical handover notes, and finalized the fully functional enterprise application ready for deployment.

## III. Technical Skills and Tools Acquired

- **Low-Code/No-Code Development:** Proficient use of Microsoft Power Apps (Canvas Apps, formula writing, galleries, form controls, and screen navigation).
- **Cloud Database Management:** Setting up, structuring, and maintaining SharePoint Lists as robust relational backend datasets.
- **Workflow Automation:** Building automated triggers and scheduled email alerts via Power Automate and Microsoft Outlook integration.
- **System UI/UX Design:** Designing intuitive navigation flows, interactive dashboards, and multi-criteria filter modules for enterprise reporting.

### IV. Problems Encountered and Solutions Implemented

| Problem / Obstacle Encountered | Root Cause | Corrective Action & Solution |
|---|---|---|
| **Data Recording Errors (Add/Edit Screens):** Initial data submissions failed to write properly to the backend source. | Improper data connection mapping and missing required field validations in form cards. | Reconfigured the data source connections, updated patch formulas, and implemented strict field validation rules. |
| **Checkbox Logic & State Management:** Checkboxes failed to retain or pass correct boolean states during record updates. | Default items and reset properties conflict in Power Apps form components. | Adjusted control formulas, updated Default property bindings, and tested form state behaviors thoroughly. |
| **Workflow Trigger Delays:** Automated notifications did not fire consistently at the scheduled time. | Incorrect time-zone configuration and trigger condition mismatches in Power Automate flows. | Corrected the recurrence schedule settings, adjusted time zones, and successfully executed manual test triggers. |

## SUMMARY, EXPERIENCE GAINED, AND SUGGESTIONS

### I. Comprehensive Summary of Internship Experience

The summer internship conducted at Airports of Thailand PLC (AOT) within the **Technology and Communications Strategy Department (ฝ่ายกลยุทธ์เทคโนโลยีและการสื่อสาร)** spanned a total duration of 37 working days, accumulating 333 productive working hours from June 1, 2026, to July 31, 2026. The core responsibility entrusted during this period involved the complete life-cycle development, integration, testing, and deployment of an enterprise-level tracking and reporting application utilizing

low-code platforms, specifically **Microsoft Power Apps**, **SharePoint Lists**, and **Power Automate**.

The internship was structured into progressive phases, moving from foundational orientation and operational site visits at Suvarnabhumi Airport to database structuring, multi-screen user interface design, automated workflow configurations, and comprehensive executive dashboard reporting. Successfully navigating this trajectory provided deep, first-hand exposure to the complexities of digital transformation within a large-scale aviation infrastructure enterprise, effectively bridging theoretical engineering concepts with practical software engineering methodologies.

## II. Detailed Experience Gained and Knowledge Application

### *A. Technical Skill Development and Low-Code Architecture*

1. **Microsoft Power Apps Development (Canvas Apps):**
    - Gained advanced practical expertise in designing and structuring responsive Canvas applications.
    - Mastered the implementation of core UI components, including multi-screen navigation, gallery controls, dynamic drop-down menus, form cards, and custom check-box logic.
    - Utilized collection variables and patch functions to manage data transactions smoothly between user interfaces and cloud databases.
2. **Cloud Database Management & Backend Integration:**
    - Acquired hands-on competence in setting up, structuring, and maintaining **SharePoint Lists** as reliable, secure relational backend data sources.
    - Learned optimal data schema design, field mapping, and data type validation to prevent data corruption and ensure seamless synchronization across multiple application screens (Add, Edit, and view modules).

3. **Workflow Automation & Real-Time Event Triggering:**
    - Leveraged **Power Automate** to design and execute robust automated workflows supporting operational needs.
    - Configured automated email notifications, user log-in alerts, and scheduled daily activity reports dispatched to Microsoft Outlook at 17:30, eliminating manual administrative overhead.
4. **Enterprise Business Intelligence & Dashboard Design:**
    - Designed and deployed multiple analytical reporting modules to transform raw transactional data into actionable executive insights.
    - Built specialized reporting views, including the Transaction List (รายการ) screen, Application Portfolio Report, Business Alignment Report, Data & PDPA Compliance Report, Infrastructure Report, Digital Technology Report, and Security / GRC Report, complete with multi-criteria filtering capabilities.

*B. Application of Academic and Engineering Principles*

- **Systematic Problem Decomposition:** Applied structured engineering problem-solving methodologies to break down complex corporate administrative workflows into modular, maintainable software requirements.
- **User-Centric UI/UX Design:** Implemented rigorous user experience principles to streamline data entry forms, minimize user input errors, and enhance the overall usability of enterprise tools for non-technical administrative staff.
- **Professional Standards & Documentation:** Cultivated rigorous technical documentation habits, maintaining precise daily logs of code modifications, formula logic, and system architectures.

## III. Suggestions and Recommendations

### *A. Recommendations for Future Application Enhancements*

1. **Role-Based Access Control (RBAC):** Future versions of the application should incorporate granular security permissions, restricting administrative editing and sensitive reporting views based on user department tiers.
2. **Advanced Data Analytics & Power BI Integration:** While the built-in Power Apps reports provide excellent operational visibility, connecting the SharePoint backend directly to **Power BI** would enable deeper predictive analytics, trend forecasting, and advanced visual dashboards for executive leadership.
3. **Automated Error Logging:** Implementing a centralized error-logging list in SharePoint would allow developers to track runtime exceptions and user-side glitches automatically.

### *B. Recommendations for Future Interns*

1. **Proactive Communication:** Maintain active, open channels of communication with department supervisors and project stakeholders early in the development cycle to clarify ambiguous requirements and prevent rework.
2. **Rigorous Version Control & Testing:** Always perform comprehensive unit and integration testing in a test environment before pushing updates to production datasets to safeguard operational data consistency.
3. **Comprehensive Knowledge Transfer:** Maintain detailed documentation of custom formulas, workflow triggers, and data structures to ensure seamless project handover for subsequent interns or department staff.